

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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2.1 Job Profile Data

The first dataset consists of all publicly available professional LinkedIn profiles between January 2021 through June 2025. The dataset includes approximately 75 million employment spells from 23 million unique individuals across 2.5 million UK-based firms. Each observation includes the worker’s job title, employer, start and end dates (if available), seniority level,³ and compensation.⁴ The data provide near-real-time employment stocks, which enable us to track within-firm workforce composition changes over time.

From the panel of individual LinkedIn profiles, we construct a monthly firm-level panel from January 2021 through June 2025. For each firm-month, we calculate total employment as the count of unique individuals with active employment spells at a certain company in that month, and log-transform this variable. This is our main employment stock measure. We then separate total employment into junior and senior employment stocks based on Revelio seniority classifications of jobs. We also calculate average compensation as the log of the mean of reported salaries plus bonuses in a firm. We only focus on firms that remained in business throughout the sample period.

To calculate firm-level exposure scores, we use the assessment framework developed by Eloundou et al. (2024). They evaluate 19,265 detailed work activities and tasks from the O*NET 27.2 database to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. We focus on their ‘E1’ measure, which captures exposure to current LLMs, rather than LLMs combined with additional software. We do so because it is the most appropriate metric for identifying short-term labor market effects that do not require additional software development or complex organizational changes. To obtain our firm-level

³Revelio Labs determines seniority using an ensemble model that combines three components: (i) current job characteristics including title, company, and industry; (ii) employment history including tenure and previous position seniority; and (iii) individual age. These three scores are averaged to create a continuous seniority metric, which is then mapped to seven ordinal levels based on keyword benchmarks (e.g., “junior,” “senior,” “director”). The levels range from Entry Level (e.g., interns, trainees) through Junior, Associate, Manager, and Director levels, to Executive and Senior Executive levels (e.g., C-suite positions).

⁴Revelio Labs estimates compensation using a predictive model trained on publicly available visa application data, self-reported salaries, and job posting information. The model incorporates position-specific variables (job title, seniority level, company, location), individual-specific factors (tenure at company), and temporal controls (year of observation) to generate salary predictions for positions where compensation data are not directly observed.

Table 4: Effect of LLM exposure on occupation-level job postings

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.039** (0.018)	-495.483*** (135.270)	-0.011*** (0.003)	-56.802 (108.358)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.953	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level outcomes. *Job postings (log)* is the log number of new vacancies in occupation o in month t . *Salary* is the mean advertised annual compensation, *Salary (log)* is its logarithm, and *Salary (SD)* is the within-occupation standard deviation of posted salaries. *Exposure* is the occupation-level LLM exposure score from Eloundou et al. (2024). *Post* equals one for November 2022 onward. Standard errors are clustered at the occupation level. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

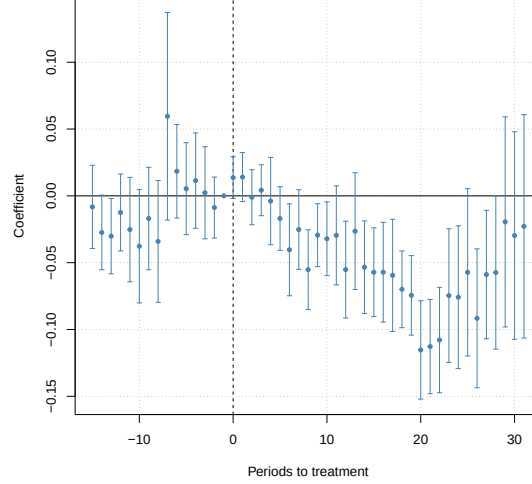
4.2.2 Occupation-Level Postings

This analysis examines occupation-level job posting data to understand how aggregate labor demand evolves for specific roles. This analysis exploits variation in LLM exposure across 384 occupations, from minimally affected roles such as cooks (exposure score: 0) to highly exposed positions like proofreaders and copy markers (exposure score: 0.86). We examine three outcomes: posting volumes to measure aggregate demand, advertised salaries to assess compensation responses, and within-occupation salary dispersion to measure wage polarization.

Figure 6 presents event study estimates for job posting volumes. Prior to November 2022, high- and low-exposure occupations follow parallel trajectories, validating our identification strategy. Following ChatGPT’s release, however, exposed occupations experience a sustained reduction in hiring demand, with effects first becoming statistically significant after 7 months and intensifying thereafter. After 20 months, we see a slight resurgence, and job postings converge back to their original levels. When averaged across the post-treatment period (Table 4), a one standard deviation increase in LLM exposure is associated with a 3.9% reduction in posting volume ($p = 0.029$). This is an economically meaningful change.

Turning to compensation, we observe immediate and persistent wage pressure for exposed occupations. Figure 7 indicates that posted salaries begin declining shortly after ChatGPT’s introduction. A one standard deviation increase in LLM exposure is associated with a £495 decline in posted salaries and a 1.1% reduction in log

Figure 6: Effect of LLM exposure on job-level hiring



Notes: This figure plots event study estimates of the effect of LLM exposure on occupation-level hiring (log vacancies per occupation-month). Coefficients are estimated from Equation (5) with occupation fixed effects and year-by-month fixed effects; standard errors are clustered at the occupation level. The dependent variable is the log of the number of new vacancies posted in occupation o in month t . Exposure is the occupation-level score from Eloundou et al. (2024). The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

salaries (both $p < 0.001$). Panel C shows that within-occupation salary dispersion remains unchanged, suggesting uniform wage pressure across the distribution rather than polarization between high- and low-earners within occupations.

These occupation-level salary declines provide a sharp contrast with the slight increase in firm-level compensation documented earlier. At the firm level, eliminating junior positions mechanically raises average compensation by shifting workforce composition toward senior roles. At the occupation level, however, we observe market-wide wage depression for exposed roles as increased labor supply (from displaced workers) meets reduced demand. As such, our results highlight the importance of considering both within-firm and within-occupation changes.

quartile serves as the reference category. For firm-level job postings, effects are significant across all higher quartiles, though the third and fourth quartiles show similar magnitudes, which is consistent with a threshold effect. For firm-level employment stocks, the evidence is weaker and non-monotonic, with only scattered significant effects across bins. The occupation-level analysis reveals clearer nonlinearities: while the second and third quartiles show minimal effects, the highest exposure quartile experiences significantly larger impacts, suggesting that most of the occupation-level results we observe are driven by the 25 percent most exposed occupations.

5 Conclusion

This paper provides evidence that large language models are reshaping UK labor markets. Using a difference-in-differences design that exploits variation in occupational- and firm-level LLM exposure, we conduct three analyses that show how labor markets change in response to the introduction of generative AI. First, we examine firm-level employment stocks to understand how workforce size and composition evolve within organizations. Second, we analyze firm-level hiring patterns to capture forward-looking recruitment decisions that precede employment adjustments. Third, we investigate occupation-level posting dynamics to identify economy-wide demand shifts.

Our firm-level employment analysis documents a 0.3 percent reduction in total employment per standard deviation increase in LLM exposure, though this effect only becomes statistically significant approximately 18 months after ChatGPT’s introduction. This reduction is predominantly concentrated in junior positions, which decline by 0.4 percent per standard deviation, while senior positions show a smaller, statistically insignificant 0.2 percent reduction. The resulting compositional shift reduces the fraction of junior workers by 0.1 percentage point per standard deviation, mechanically increasing average compensation by £106 as firms retain proportionally more higher-paid senior staff, though this compensation effect is only marginally significant.

While employment stocks adjust gradually, job posting data provide real-time evidence of firms’ evolving labor demand. Firm-level posting activity drops following

ChatGPT’s release, with a one standard deviation increase in LLM exposure reducing the probability of posting any vacancy by 1.0 percentage point. This reduction emerges gradually and becomes significant after approximately 10 months, mirroring the timeline of employment stock adjustments. When we examine the composition of roles driving this reduction, we find that technical positions and creative roles experience the sharpest declines. Sales representative positions, by contrast, show a statistically significant increase, suggesting that customer-facing roles requiring direct interpersonal interaction remain robust to LLM-based automation.

In addition to within-firm dynamics, our occupation-level analysis reveals how labor demand shifts for entire job categories regardless of firm exposure. A one standard deviation increase in LLM exposure is associated with a 3.9 percent reduction in posting volumes, accompanied by £495 (1.1 percent) salary declines. The apparent inconsistency of declining posted salaries for exposed occupations alongside increasing firm-level compensation shows how aggregation could mask opposing forces. While LLM exposure drives down market wages for specific occupations, within-firm rebalancing toward more senior positions pushes average firm-level wages in the opposite direction.

Our heterogeneity analysis reveals that the observed labor market disruptions are almost entirely driven by high-remuneration segments of the labor market. This concentration of effects in higher-wage segments contradicts historical automation patterns that predominantly displaced routine middle-skill work, and suggests that LLMs might lead to a compression of the wage distribution.

Our findings carry important implications for understanding technological displacement and designing policy responses. The concentration of job losses among entry-level positions disrupts traditional skill development pathways where workers master progressively complex tasks through hands-on experience. Without junior roles serving as training grounds, firms may struggle to develop senior talent internally while new entrants face shrinking opportunities to gain experience (Ide, 2025). The freeze in hiring followed by employment stock changes suggests that forward-looking indicators like job postings may provide early warning signals for policymakers. The sharp reductions in technical and analytical roles alongside resilience in sales positions indicates that human interaction and relationship-building skills remain valuable complements to LLM capabilities, suggesting that retraining programs should emphasize these interpersonal competencies. Policy responses